

Lab 6 & 7 - Report

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Group: Mon 11:00-13:00

Lab 6

1. Part 1: Examine Network Requirements

Step 1.

a) $32 - 25 = 7$

$2^7 - 2 = 126$

2) $40 + 25 + 5 + 4 + 4 + 2 = 80$

3) 6 subnets

Step 2

BR1 LAN

$40 < 2^6$ $32 - 6 = 26$

$2^6 - 2 = 62$

Step 3

BR2 LAN

$25 < 2^5$ $32 - 5 = 27$

$2^5 - 2 = 30$

Step 4

4, 5 < 2^3 $32 - 3 = 29$

$2^3 - 2 = 6$

Step 5

$2 < 2^2$ $32 - 2 = 30$

$2^2 - 2 = 2$

Step 1: Determine how many host addresses and subnets are available.

How many host addresses are available in a /25 network?

126

What is the total number of host addresses needed in the topology diagram?

80

How many subnets are needed in the network topology?

6

Step 2: Determine the largest subnet.

What is the subnet description (e.g. BR1 LAN or BR1-BR2 link)?

BR1 LAN

How many IP addresses are required in the largest subnet?

40

What subnet mask can support that many host addresses?

/26 (255.255.255.192)

How many total host addresses can that subnet mask support?

62

Can you subnet the 192.168.33.128/25 network address to support this subnet?

Yes

What are the network addresses that would result from this subnetting?

192.168.33.128/26 and 192.168.33.192/26

Step 3: Determine the second largest subnet.

What is the subnet description?

BR2 LAN

How many IP addresses are required for the second largest subnet?

25

What subnet mask can support that many host addresses?

/27 or 255.255.255.224

How many total host addresses can that subnet mask support?

30

Can you subnet the remaining subnet again and still support this subnet?

Yes

What are the network addresses that would result from this subnetting?
192.168.33.192/27 and 192.168.33.224/27

Step 4: Determine the third largest subnet.

What is the subnet description?

BR2 IoT LAN

How many IP addresses are required for the next largest subnet?

5

What subnet mask can support that many host addresses? /29 or
255.255.255.248

How many total host addresses can that subnet mask support?

6

Can you subnet the remaining subnet again and still support this subnet?
yes

What are the network addresses that would result from this subnetting?
192.168.33.224/29, 192.168.33.232/29, 192.168.33.240/29, 192.168.33.248/29

Step 5: Determine the fourth largest subnet.

What is the subnet description?

BR11-BR2 Link

How many IP addresses are required for the next largest subnet?

2

What subnet mask can support that many host addresses?
/30 or 255.255.255.252

How many total host addresses can that subnet mask support?

2

Can you subnet the remaining subnet again and still support this subnet?
yes

What are the network addresses that would result from this subnetting?
192.168.33.248/30, 192.168.33.252/30

2. Part 1: Examine Network Requirements

Part 2

1) BR1 LAN (40, 2^6)

$$32 - 6 = 26$$

$$128 + 1 = 129$$

$$128 + 64 - 1 = 191$$

2) BR2 ~~LAN~~ (25, 2^5)

$$32 - 5 = 27$$

$$192 + 1 = 193$$

$$192 + 32 - 1 = 223$$

3) BR2 ~~LAN~~ IoT (5, 2^3)

$$32 - 3 = 29$$

$$224 + 1 = 225$$

$$224 + 8 - 1 = 231$$

4) BR2 CCTV (4, 2^3)

$$32 - 3 = 29$$

$$232 + 1 = 233$$

$$232 + 8 - 1 = 239$$

5) BR2 HVAC (4, 2^3)

$$32 - 3 = 29; \quad 240 + 1 = 241; \quad 240 + 8 - 1 = 247$$

6) BR1 - BR2 Link (2, 2^4)

$$32 - 2 = 30; \quad 248 + 1 = 249$$

$$248 + 4 - 1 = 251$$

Step 1: Calculate the subnet information.

Subnet Description	Number of Hosts Needed	Network Address /CIDR	First Host Address	Broadcast Address
BR1 LAN	40	192.168.33.128/25	192.168.33.129	192.168.33.191
BR2 LAN	25	192.168.33.192/27	192.168.33.193	192.168.33.223
BR2 IoT LAN	5	192.168.33.224/29	192.168.33.225	192.168.33.231
BR2 CCTV LAN	4	192.168.33.232/29	192.168.33.233	192.168.33.239
BR2 HVAC C2LAN	4	192.168.33.240/29	192.168.33.241	192.168.33.247
BR1-BR2 Link	2	192.168.33.248/30	192.168.33.249	192.168.33.251

Step 2: Complete the device interface address table.

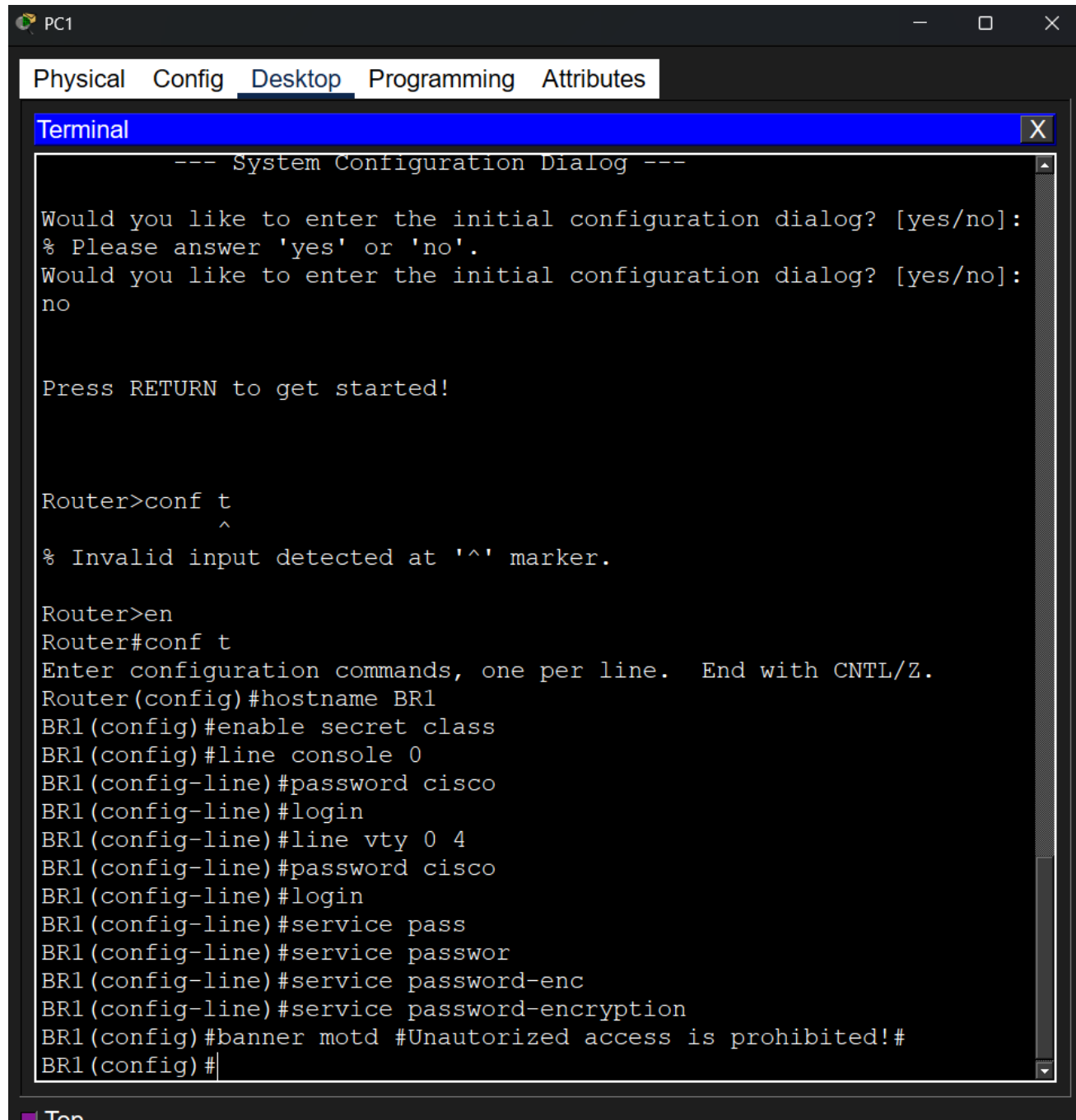
Device	Interface	IP Address	Subnet Mask	Device Interface
BR1	G0/0/0	192.168.33.249	255.255.255.252	BR1-BR2 Link
	G0/0/1	192.168.33.129	255.255.255.192	40 Host LAN
BR2	G0/0/0	192.168.33.250	255.255.255.252	BR1-BR2 Link
	G0/0/1	192.168.33.193	255.255.255.224	25 Host LAN

3. Part 3: Cable and Configure the IPv4 Network

Step 1



Step 2



The screenshot shows a terminal window titled 'PC1' with tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Terminal' tab is active, displaying a 'System Configuration Dialog'. The user is prompted to enter the initial configuration dialog, which they decline. They then enter 'conf t' at the Router prompt, which is rejected as invalid. After entering 'en' to enter configuration mode, they proceed with several commands to configure the router: setting the hostname to 'BR1', enabling secret class, configuring console and vty lines with passwords and login, setting service password encryption, and adding a MOTD banner. The terminal text is as follows:

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]:
no

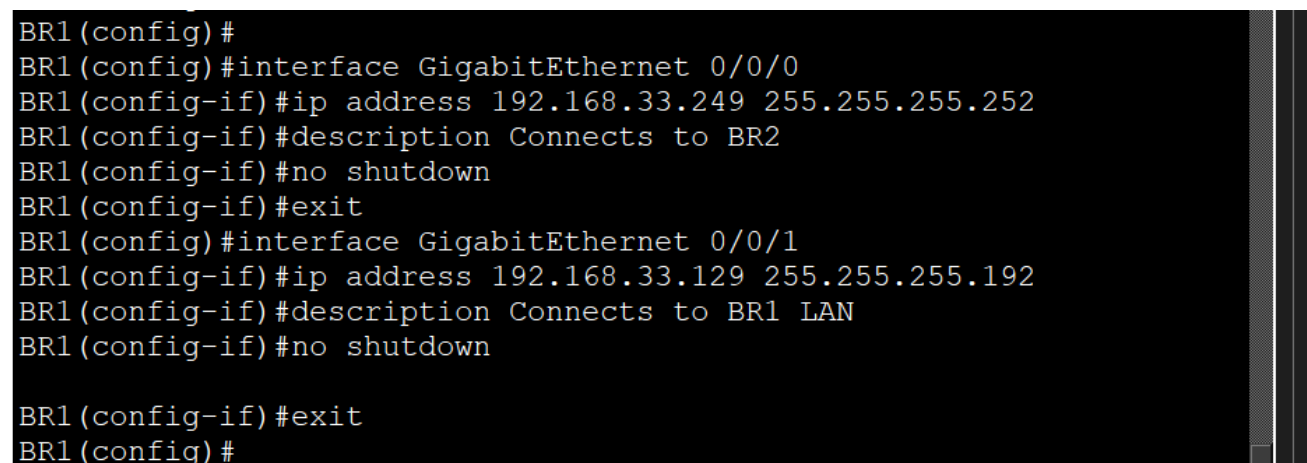
Press RETURN to get started!

Router>conf t
      ^
% Invalid input detected at '^' marker.

Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname BR1
BR1(config)#enable secret class
BR1(config)#line console 0
BR1(config-line)#password cisco
BR1(config-line)#login
BR1(config-line)#line vty 0 4
BR1(config-line)#password cisco
BR1(config-line)#login
BR1(config-line)#service pass
BR1(config-line)#service passwor
BR1(config-line)#service password-enc
BR1(config-line)#service password-encryption
BR1(config)#banner motd #Unauthorized access is prohibited!#
BR1(config)#
```

At the bottom of the terminal window, there is a 'Top' button.

Step 3



The screenshot shows a terminal window with the following configuration commands for interface GigabitEthernet 0/0/0 and 0/0/1 on router BR1:

```
BR1(config)#
BR1(config)#interface GigabitEthernet 0/0/0
BR1(config-if)#ip address 192.168.33.249 255.255.255.252
BR1(config-if)#description Connects to BR2
BR1(config-if)#no shutdown
BR1(config-if)#exit
BR1(config)#interface GigabitEthernet 0/0/1
BR1(config-if)#ip address 192.168.33.129 255.255.255.192
BR1(config-if)#description Connects to BR1 LAN
BR1(config-if)#no shutdown

BR1(config-if)#exit
BR1(config)#
```

Step 4

```
BR1#wr mem
Building configuration...
[OK]
BR1#
```

Step 5

```
BR2#
BR2#ping 192.168.33.249

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.33.249, timeout is 2
seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1
ms

BR2#
```

```
BR1#ping 192.168.33.250

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.33.250, timeout is 2
seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/3 ms

BR1#
```

Reflection Question

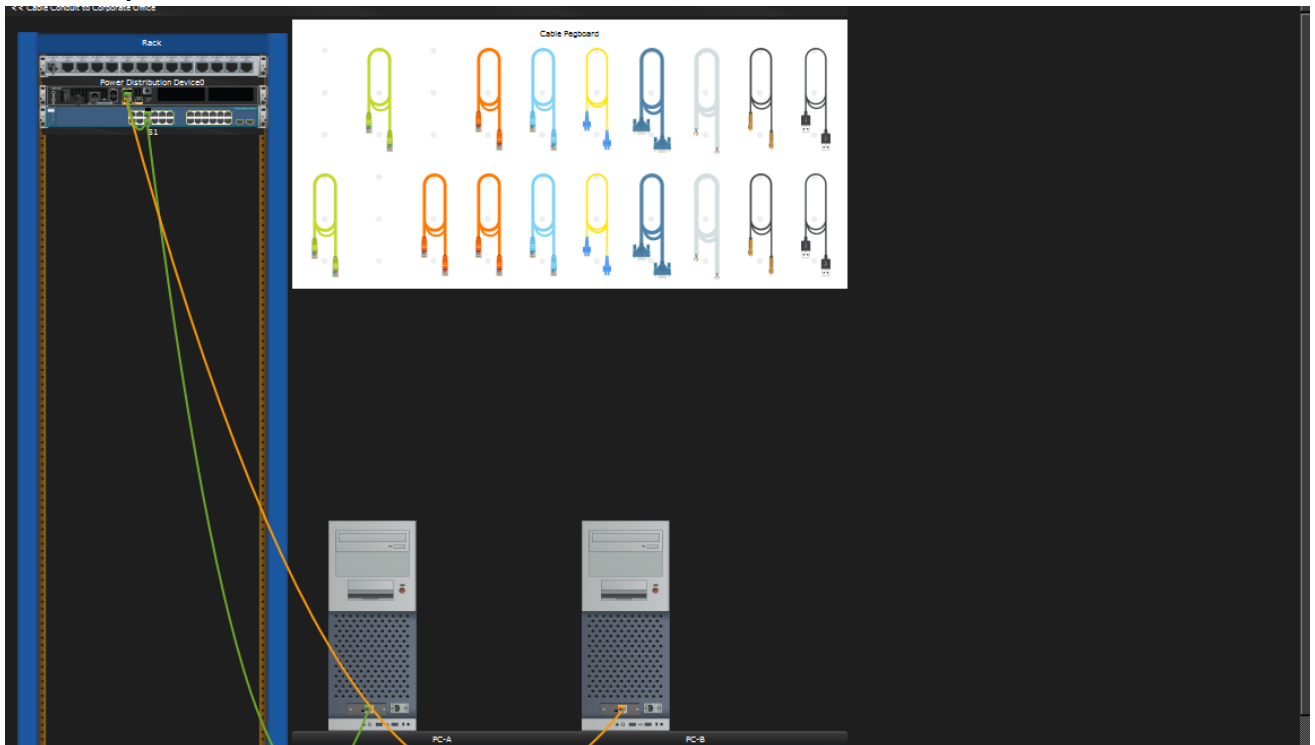
Can you think of a shortcut for calculating the network addresses of consecutive /30 subnets?

The step between network addresses in a /30 mask is always 4. To find the next /30 subnet, simply add 4 to the current network address.

Lab 7

Part 1: Cable the Network and Configure Basic Router and Switch Settings

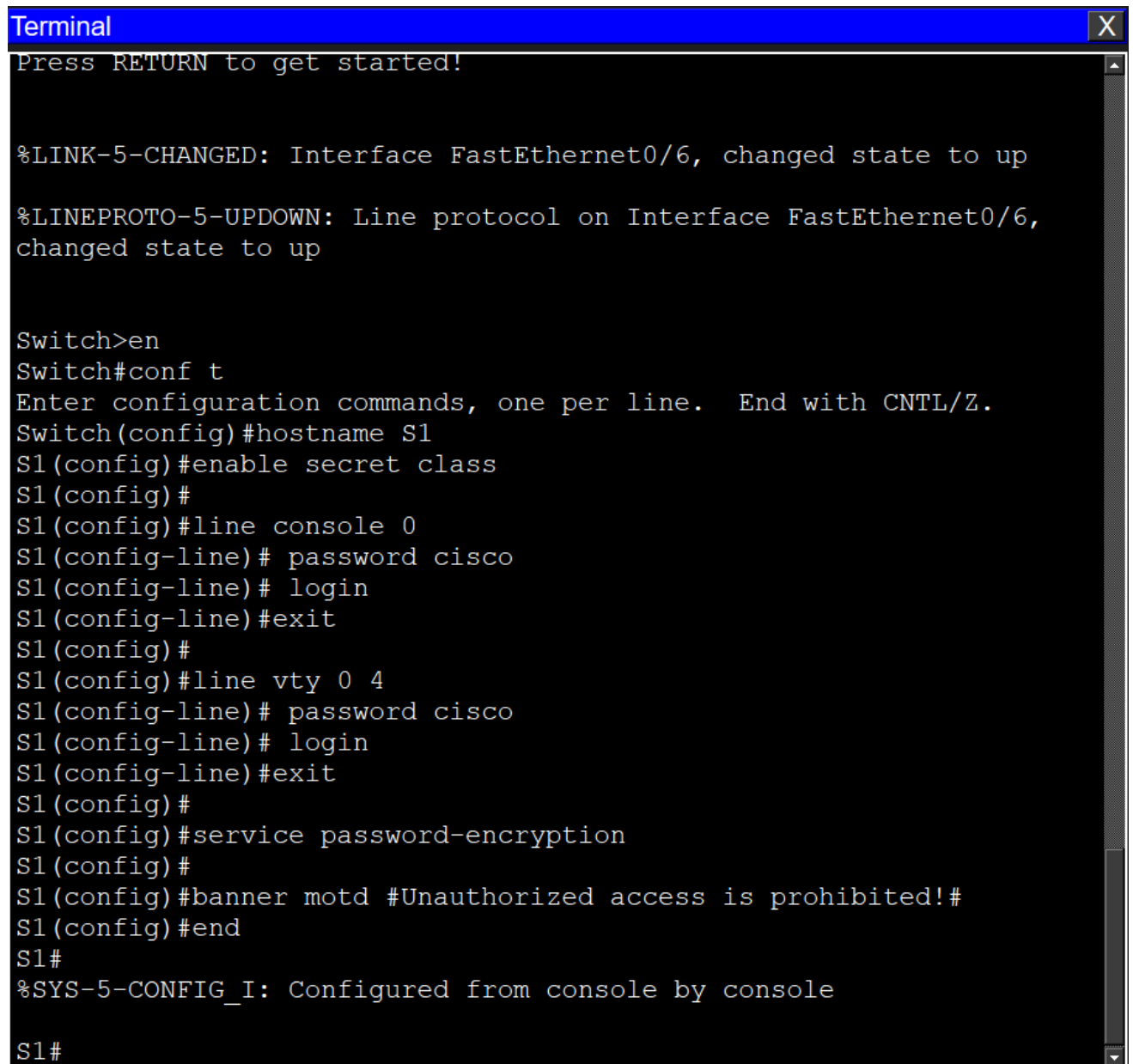
Step 1



Step 2

```
Would you like to enter the initial configuration dialog? [yes/no]:  
n  
  
Press RETURN to get started!  
  
Router>en  
Router#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#hostname R1  
R1(config)#enable secret class  
R1(config)#  
R1(config)#line console 0  
R1(config-line)# password cisco  
R1(config-line)# login  
R1(config-line)#exit  
R1(config)#  
R1(config)#line vty 0 4  
R1(config-line)# password cisco  
R1(config-line)# login  
R1(config-line)#exit  
R1(config)#  
R1(config)#service password-encryption  
R1(config)#  
R1(config)#banner motd #Unauthorized access is prohibited!#  
R1(config)#  
R1(config)#end  
R1#  
%SYS-5-CONFIG_I: Configured from console by console
```


Step 3



```
Terminal
Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up

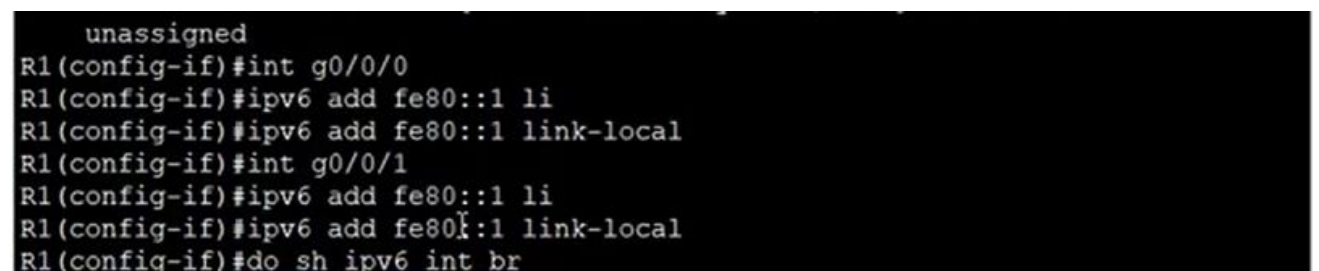
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6,
changed state to up

Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#enable secret class
S1(config)#
S1(config)#line console 0
S1(config-line)# password cisco
S1(config-line)# login
S1(config-line)#exit
S1(config)#
S1(config)#line vty 0 4
S1(config-line)# password cisco
S1(config-line)# login
S1(config-line)#exit
S1(config)#
S1(config)#service password-encryption
S1(config)#
S1(config)#banner motd #Unauthorized access is prohibited!#
S1(config)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#
```

Part 2: Configure IPv6 Addresses Manually

Step 1



```
unassigned
R1(config-if)#int g0/0/0
R1(config-if)#ipv6 add fe80::1 li
R1(config-if)#ipv6 add fe80::1 link-local
R1(config-if)#int g0/0/1
R1(config-if)#ipv6 add fe80::1 li
R1(config-if)#ipv6 add fe80::1 link-local
R1(config-if)#do sh ipv6 int br
```

What is the multicast group for the G0/0/0 interface?

You will see groups FF02::1 and FF02::1:FF00:1. These groups allow nodes on the local network to find each other.

```

R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int g0
R1(config)#int g\
R1(config)#int g0/0
R1(config)#int g0/0/0
R1(config-if)#ipv6 address 2001:db8:acad:1::1/64
R1(config-if)#ipv6 address fe80::1 link-local
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0/0, changed state to up

R1(config-if)#interface GigabitEthernet 0/0/1
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
%GigabitEthernet0/0/1: Error: 2001:DB8:ACAD:1::/64 is overlapping
with 2001:DB8:ACAD:1::/64 on GigabitEthernet0/0/0
R1(config-if)# ipv6 address fe80::1 link-local
R1(config-if)# no shutdown

R1(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface

```

PC-A

Physical Config Desktop Programming

Terminal

```

R1(config)#sh ipv6 int br
^
% Invalid input detected at '^' marker.

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#sh ipv6 int br
GigabitEthernet0/0/0      [up/up]
    FE80::1
    2001:DB8:ACAD:1::1
GigabitEthernet0/0/1      [up/up]
    FE80::1
Vlan1                    [administratively down/down]
    unassigned

R1#sh ipv6 int g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::1
  No Virtual link-local address(es):
  Global unicast address(es):
    2001:DB8:ACAD:1::1, subnet is 2001:DB8:ACAD:1::/64
  Joined group address(es):
    FF02::1
    FF02::1:FF00:1
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds

R1#

```

Step 2

```
S1>
S1>en
Password:
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#ipv6 unicast-routing
S1(config)#exit
S1#
%SYS-5-CONFIG_I: Configured from console by console
```

Has an IPv6 unicast address been assigned to the network interface card (NIC) on PC-B?

Answer: No. (Before we enabled routing on the router, the PC could only have a Link-Local address starting with fe80::).

Why did PC-B receive the Global Routing Prefix and Subnet ID that you configured on R1?

Because router R1 started sending RA messages. PC-B, using the SLAAC mechanism, automatically extracted the network prefix from this message and generated a fully functional address.

Step 3

```
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#interface vlan 1
S1(config-if)#ipv6 address 2001:db8:acad:1::b/64
S1(config-if)#ipv6 address fe80::b link-local
S1(config-if)#no shutdown

S1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed
```

Step 4

IPv6 Configuration

☒ Automatic
☒ Static

IPv6 Address
2001:db8:acad:1::3 / 64

Link Local Address
FE80::204:9AFF:FEEA:9CBC

Default Gateway
FE80::1

DNS Server

- 802 1X

IPv6 Configuration

☒ Automatic
☒ Static

IPv6 Address
2001:db8:acad:a::3 / 64

Link Local Address
FE80::2D0:58FF:FE51:5E24

Default Gateway
FE80::1

DNS Server

Part 3: Verify End-to-End Connectivity

```

C:\>ping fe80::1

Pinging fe80::1 with 32 bytes of data:

Reply from FE80::1: bytes=32 time<1ms TTL=255
Reply from FE80::1: bytes=32 time<1ms TTL=255
Reply from FE80::1: bytes=32 time<1ms TTL=255
Reply from FE80::1: bytes=32 time<1ms TTL=255

Ping statistics for FE80::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>tracert 2001:db8:acad:a::3

Tracing route to 2001:db8:acad:a::3 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      2001:DB8:ACAD:1::1
  2  0 ms      0 ms      0 ms      2001:DB8:ACAD:A::3

Trace complete.

C:\>

```

```
C:\>ping 2001:db8:acad:1::3

Pinging 2001:db8:acad:1::3 with 32 bytes of data:

Reply from 2001:DB8:ACAD:1::3: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:1::3: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:1::3: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:1::3: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:ACAD:1::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping fe80::1

Pinging fe80::1 with 32 bytes of data:

Reply from FE80::1: bytes=32 time<1ms TTL=255
Reply from FE80::1: bytes=32 time<1ms TTL=255
Reply from FE80::1: bytes=32 time<1ms TTL=255
Reply from FE80::1: bytes=32 time<1ms TTL=255

Ping statistics for FE80::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Reflection Question

1. Why can the same link-local address, fe80::1, be assigned to both Ethernet interfaces on R1?

Packets with link-local addresses never leave their local network. Since interfaces G0/0/0 and G0/0/1 point to different isolated networks, there is no address conflict.

2. What is the Subnet ID of the IPv6 address 2001:db8:acad::aaaa:1234/64, if the network prefix is /48?

0000. With a /48 prefix, the global part is 2001:db8:acad. The fourth hextet is the Subnet ID. In this address, after acad, there are double colons ::, indicating padding with zeros.